—: Genetic Algorithm :-

27/1/2026

structure :-

Population initialization
↓
→ fitness assignment
↓
selection
↓
crossover
↓
mutation
↓
survivor selection
↓
terminate & return best

GA operators

Knapsack problem by using Genetic Algorithm :-

Item	weight	Value
A	5 kg	$12
B	3 kg	$5
C	7 kg	$10
D	2 kg	$7

Knapsack(K)
K=12 kg

step 1 :- chromosomes encoding

	A	B	C	D
C_i				

Gene :- 0 → represents absence of item in the knapsack
1 → represents presence of item in the knapsack.

4 bits are required to represent chromosomes encoding

set space = 2^4

Initial population is created and chromosomes randomly created.

C_1	0	1	1	0
C_2	0	1	0	1
C_3	1	1	0	1
C_4	1	1	1	1

randomly

step2 :- fitness function

(i) $C_1 \rightarrow$ 0 1 1 0
A B C D

value of Knapsack = B + C = 5 + 10 = 15
weight of Knapsack = B + C = 3 + 7 = 10
C1 accepted.

(ii) C2 $\rightarrow$ 0 1 0 1
A B C D

Value = B + D = 5 + 7 = 12 accepted
weight = B + D = 3 + 2 = 5

(iii) C3 $\rightarrow$ 1 1 0 1
A B C D

Value = A + B + D = 12 + 5 + 10 = 24 accepted
weight = A + B + D = 5 + 3 + 2 = 10

(iv) C4 $\rightarrow$ 1 1 1 1
A B C D

Value = A + B + C + D = 12 + 5 + 10 + 7 = 34
weight = 5 + 3 + 7 + 2 = 17 kg rejected.

step3 :- selection

Roulette wheel selection process

-> spin the roulette wheel and whenever the wheel stops, the individual gets selected at that point.

-> The individual that has the highest fitness value gets larger share of the wheel.

total fitness value = 15 + 12 + 24 + 0 = 51

C3 occupies half of the wheel as 24/51.

next generation :- C3, C2, C1

C3
C1
C3
C2

no of things to be selected = Valū × n (no of gen) step 4 :- cross over

C3 :- 1 1 0 | 1 → C3' :- 1 1 0 0

C1 :- 0 1 1 | 0 → C1' :- 0 1 1 1

next gener :- C3'
C1'
C3
C2

step 5 :- mutation → diversity within population so → no stuck at local minima

C :- 0 1 1 0

after mut :- 0 1 0 0 } diversity

2/2/2026

Evaluation of models: - Decision tree / Baseline model

Comparing two models A & B → NN

compute accuracy on the two models.

Input / Batch / fold	Acc(A)	Acc(B)	Diff(x)
1	.7	.75	0.05
2	.68	.72	0.04
3	⋮	⋮	⋮
⋮			
K fold	.71	.70	− 0.01

Test whether model B is statistically better than model A?

Hypothesis: H1: model B is statistically better than "A".

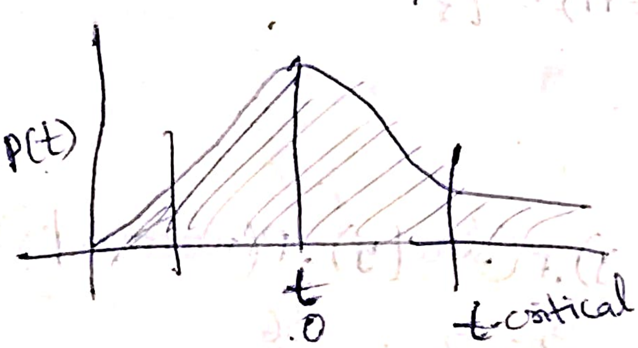
Null hypothesis :- H0: model B is not better than A.

$$t\text{-statistic} = \frac{\mu(x)}{\sigma(x)/\sqrt{n}}$$

$$\frac{0.05+0.04-0.01}{3} = 0.026$$

$$\sqrt{\frac{(0.05-0.026)^2+(0.04-0.026)^2+(0.026+0.01)^2}{3-1}}$$

$$= 0.032$$



$$t = \frac{0.026}{0.032} \cdot \sqrt{3}$$

$$= 1.407$$

|t-stat| < t-critical

P → area under curve from

signifi → " " " from t_{crit} to ∞

~~Assign~~:

9/2/2026

Bayesian learning:-

→ Probabilistic framework for reasoning about hypothesis in presence of uncertainity

→ It considers not a single hypothesis but the distribution of the hypothesis.

ws	outlook	temp	Humidity	wind	play
2.5	Sunny	High	high	strong	n
3.5	Sunny	low	high	weak	n
3.1	overcast	high	high	weak	y ✓
3.4	Sunny	low	normal	weak	y ✓
3.2	Sunny	high	normal	strong	y ✓
3	sunny	high	normal	strong	n
2.8	rainy	high	normal	weak	y ✓
2.0	rainy	low	high	strong	y ✓

Bayes theorem:-

$$P(y/x) = \frac{\overbrace{P(x/y)}^{\text{likelyhood}} \cdot \underbrace{P(y)}_{\leftarrow\text{prior}}}{P(x)}$$

Posterior ↗

$P(x)$ ← total proba

⇓

$P(y=y)P(x/y=y) + P(y=n)P(x/y=n)$

prior probabilty:- (y,n)

$P(y=y) = 5/8 \qquad P(y=n) = 3/8$

$P(x/y=y) = 1/5$

⇓

$$\prod_{i=1}^{4} (P(x_i/y=y) = P(\text{sunny}/y).P(\text{low}/y).P(\text{normal}/y).P(\text{weak}/y)$$

$$= \frac{2}{5}\cdot\frac{2}{5}\cdot\frac{3}{5}\cdot\frac{3}{5} = \frac{36}{625}$$

$$\boxed{P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}$$

$y^* = \max_{y\in\{0,1\}} \left\{ \begin{matrix} P(y=1/x)\ (\hat{\theta}1) \\ P(y=0/x) \end{matrix} \right\}$

Naive Bayes classification model :-

Compute priors, $P(y)$, $\forall\ y \in \{1,0\}$

Compute likelihood, $P(x_i|y_i)\ \forall\ x_i \in D$
$y \in \{1,0\}$

outlook :-	$P(\text{sunny}/y=1) = \frac{2}{5}$	$P(\text{overcast}/y=1) = 1/5$	$P(\text{rainy}/y=1) = 2/5$
	$P(\text{sunny}/y=0) = 3/3$	$P(\text{overcast}/y=0) = 0$	$P(\text{rainy}/y=0) = 0$
temp :-	$P(\text{low}/y=1) = 2/5$	$P(\text{high}/y=1) = 3/5$	
	$P(\text{low}/y=0) = 1/3$	$P(\text{high}/y=0) = 2/3$	
humid :-	$P(\text{high}/y=1) = 2/5$	$P(\text{normal}/y=1) = 3/5$	
	$P(\text{high}/y=0) = 2/3$	$P(\text{normal}/y=0) = 1/3$	
wind :-	$P(\text{strong}/y=1) = 2/5$	$P(\text{weak}/y=1) = 3/5$	
	$P(\text{strong}/y=0) = \frac{2}{3}$	$P(\text{weak}/y=0) = 1/3$	

let $x = \{\text{sunny, low, high, weak}\}$

$y^* =$ $y=0$: $P(\text{sunny}/y)\ P(\text{low}/y)\ P(\text{high}/y)\ P(\text{weak}/y)$

$= \frac{3}{3} \times \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} \Rightarrow \frac{6}{81} = 0.0740$

$y=1$: $P(\text{sunny}/y)\ P(\text{low}/y)\ P(\text{high}/y)\ P(\text{weak}/y)$

$= \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} = \frac{24}{625} = 0.0384$

$$P(y=1/x) = \underbrace{P(x/y=1)}_{\text{likelihood}}\ \underbrace{(P(y=1))}_{\text{Prior}} \Rightarrow \frac{24}{625} \times \frac{5}{8} = 0.024$$

$$P(y=0/x) = P(x/y=0)\ P(y=0) \Rightarrow \frac{6}{81} \times \frac{3}{8} = 0.027$$

wind speed :- $x =$
$\mu_1 =$
$\sigma_2 =$

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \rightarrow y=1$$

$\mu_2, \sigma_2 = y=0$ =

* probability of a single trail with Binary outcome $y = \{0, 1\}$ follows Bernouli distribution, given the overall Prior Probability p.

$$P(y) = p^{y}(1-p)^{1-y}$$

<u>Binomial</u> <u>distribution</u> :- probability of getting r success out of n trials repeated.

$$P(r) = ({}^{n}C_{r})\, p^{r}(1-p)^{n-r}$$

fair coin 5 times tossed

Ex :- $\dfrac{5!}{3! \times 2!}\left(\dfrac{1}{2}\right)^{3}\left(\dfrac{1}{2}\right)^{2} \Rightarrow \dfrac{5 \times \cancel{4}^{2} \times \cancel{3} \times \cancel{2} \times \cancel{1}}{\cancel{3} \times \cancel{2} \times 1 \times \cancel{2} \times \cancel{1}}\left(\dfrac{1}{2}\right)^{5}$

$$= 10 \times \frac{1}{32} = \underline{0.3125}$$

$$P(y/n) = E[y] = \max \text{ likelihood}$$
$$= \max \prod_i P(x_i | y)$$

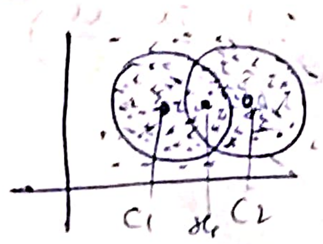
<u>Maximum</u> <u>likelihood</u> <u>Estimator</u> :- (MLE)

$$P(y|x) = \max \prod_i P(x_i | y)$$

mle -> binomial dn -> find expected value

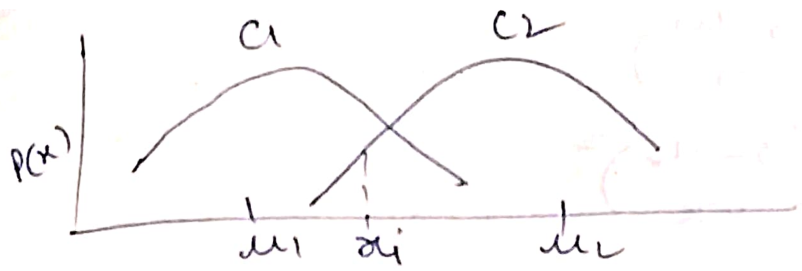
$$\log(P(y/n)) = \sum_i \log(P(x_i|y))$$

<u>Estimating</u> parameters in <u>Gaussian mixture model</u> (GMM) using Expectation <u>maximization</u> (EM) method :-



$$\left.\begin{array}{l} x_i \in \{C_1, C_2\} \\ x_i \in \{1, 0\} \end{array}\right\} \text{Hard assignment}$$

$x_i \in \{0.8, 0.2\}$ -> soft assign



let the a hidden component (Indicator variable)

$$z_{ij} = \begin{cases} 1 & \text{if } x \in j^{th} \text{ Gaussian} \\ 0 & \text{else} \end{cases}$$

soft member

E-step: $\hat{\gamma}_{ij} = E[z_{ij}] = \dfrac{\pi_j N(x_i \mid \mu_j, \sigma_j^2)}{\sum \pi_k N(x_i \mid \mu_k, \sigma_k^2)}$

m-step: $\pi_j = \dfrac{\sum \gamma_{ij}}{n}$, $\mu_j = \dfrac{\sum \gamma_{ij} x_i}{\sum \gamma_{ij}}$

EM algorithm:-

① Initialize π_j, μ_j, σ_j

② E - step

③ m - step

④ Repeat step 2 & 3 until convergence.

$X = \{2, 3, 8, 5, 4, 1, 10, 9, 6\}$

let there are 2 gaussian components/distributions.

Initialize : $\mu_1 = \mu_2 = 0.5$, $\sigma_1 = \sigma_2 = 1$, $\pi_1 = \pi_2 = 0.5$ ✗ $\mu_1 = 3, \mu_2 = 6$, $\sigma_1 = \sigma_2 = 1$, $\pi_1 = \pi_2 = 0.5$

$i \longrightarrow$

γ_{ij} $j \downarrow$

1	2	3	4	5	6	7	8	9
	$\frac{1}{2}$							

$$\gamma_{12} = \frac{0.5 \frac{1}{\sqrt{2\pi} \, 1} e^{-\frac{1}{2}\left(\frac{3-0.5}{1}\right)^2}}{0.5 \frac{1}{\sqrt{2\pi}} \sum e^{-\frac{1}{2}\left(\frac{x_i - 0.5}{1}\right)^2}}$$

2.25
6.25
56.25
20.25
12.25
0.25
90.25
72.25